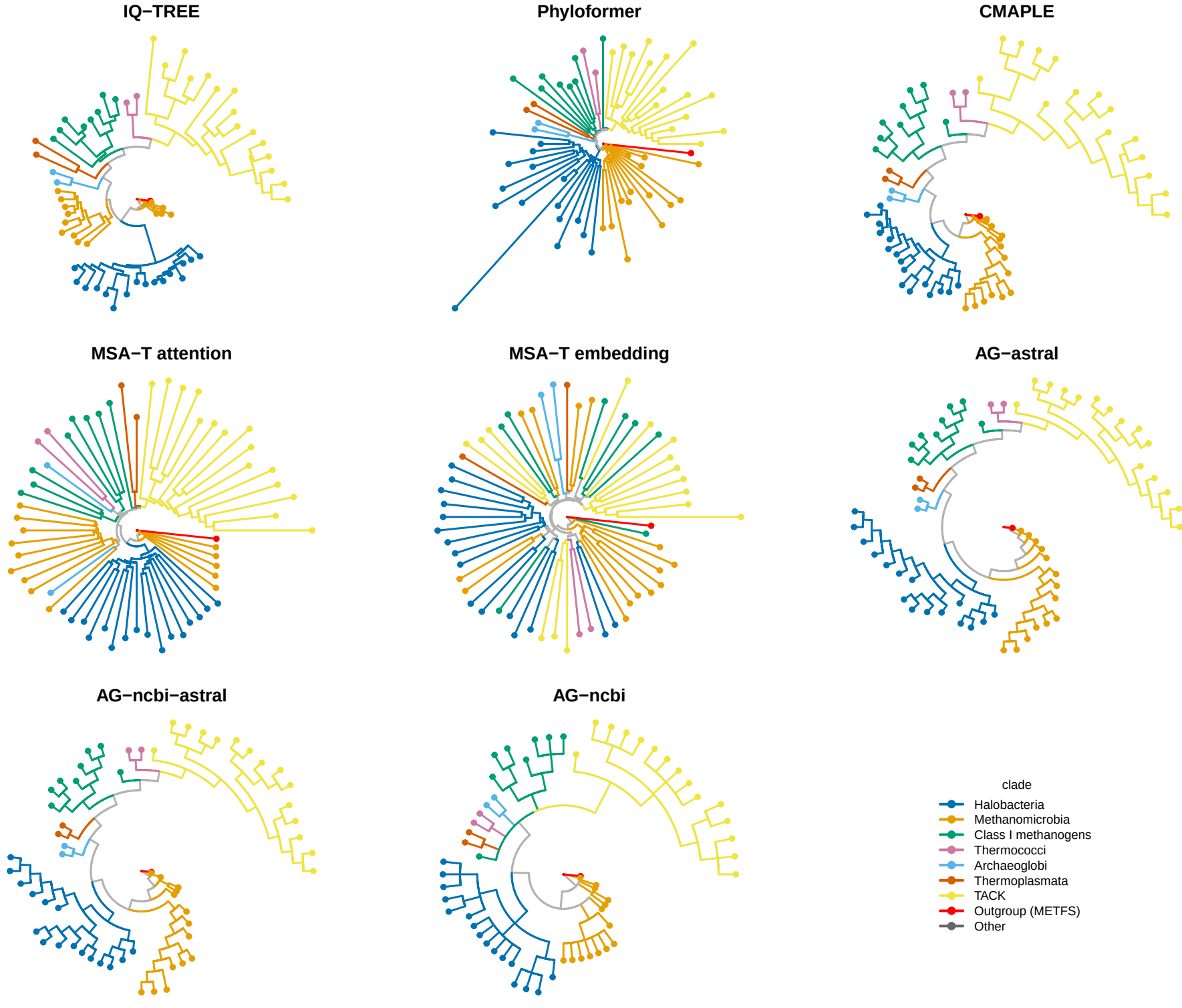
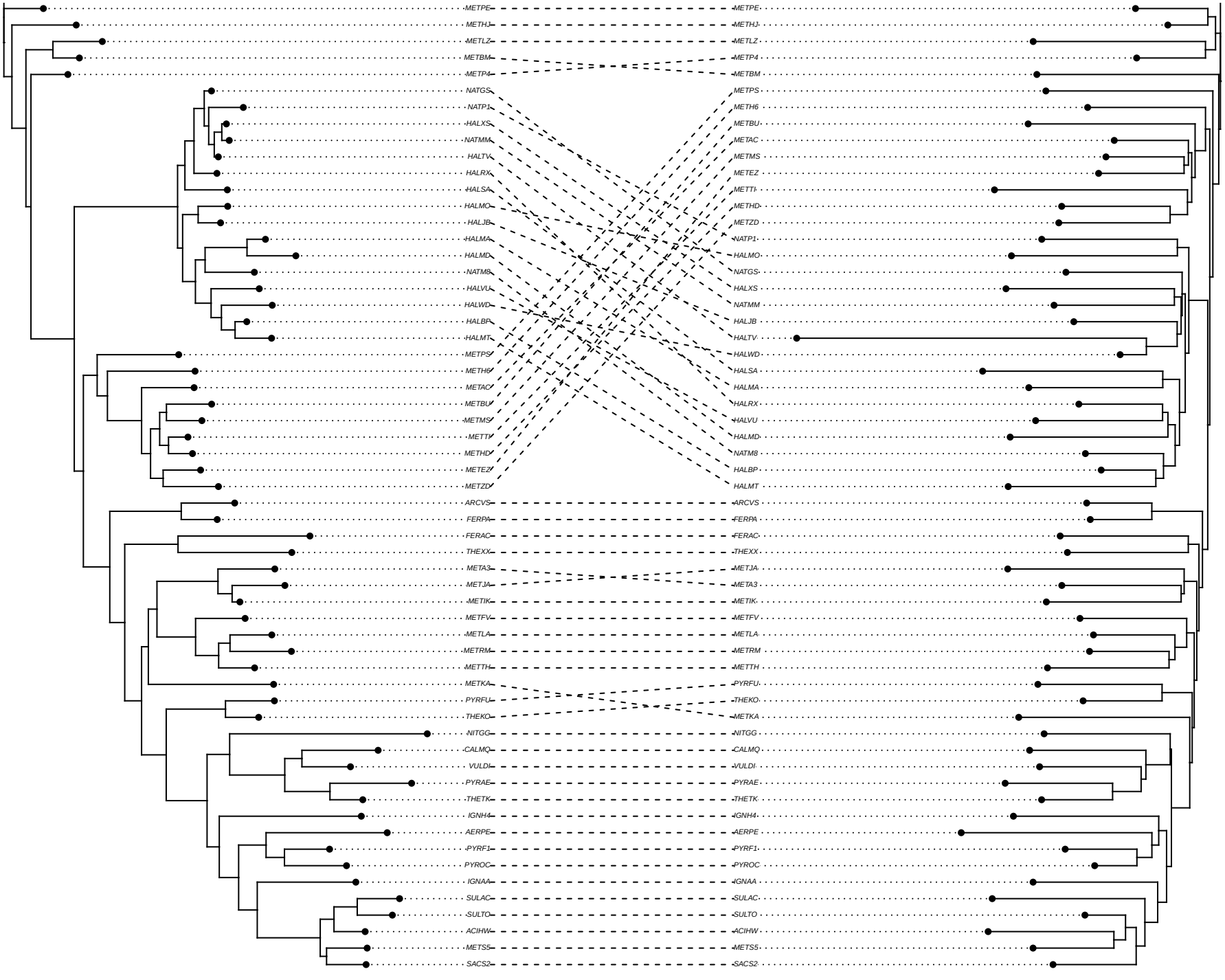
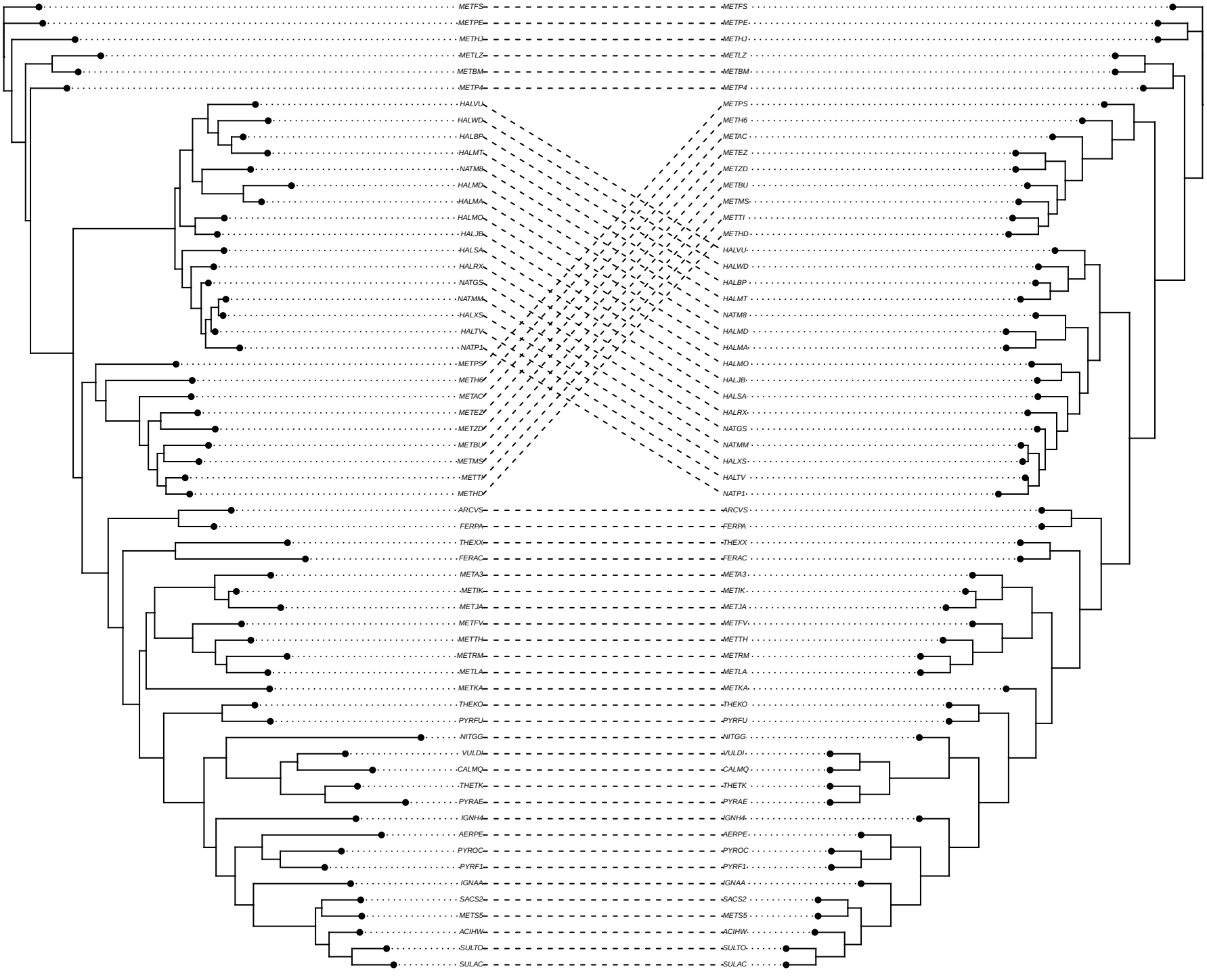

IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
AG−astral
AG−ncbi−astral
AG−ncbi
clade
Halobacteria
Methanomicrobia
Class I methanogens
Thermococci
Archaeoglobi
Thermoplasmata
TACK
Outgroup (METFS)
Other

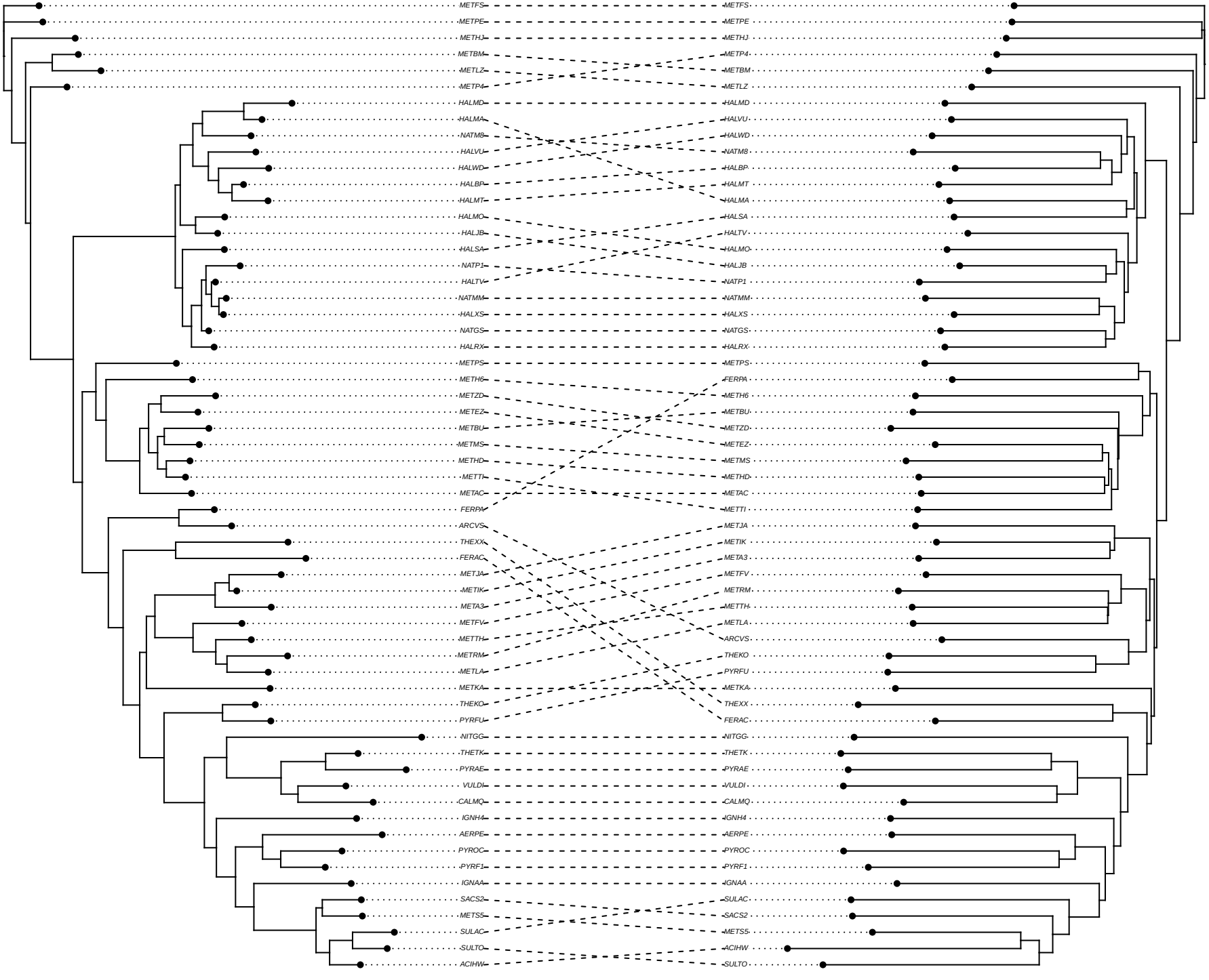
# IQ-TREE vs Phylolformer



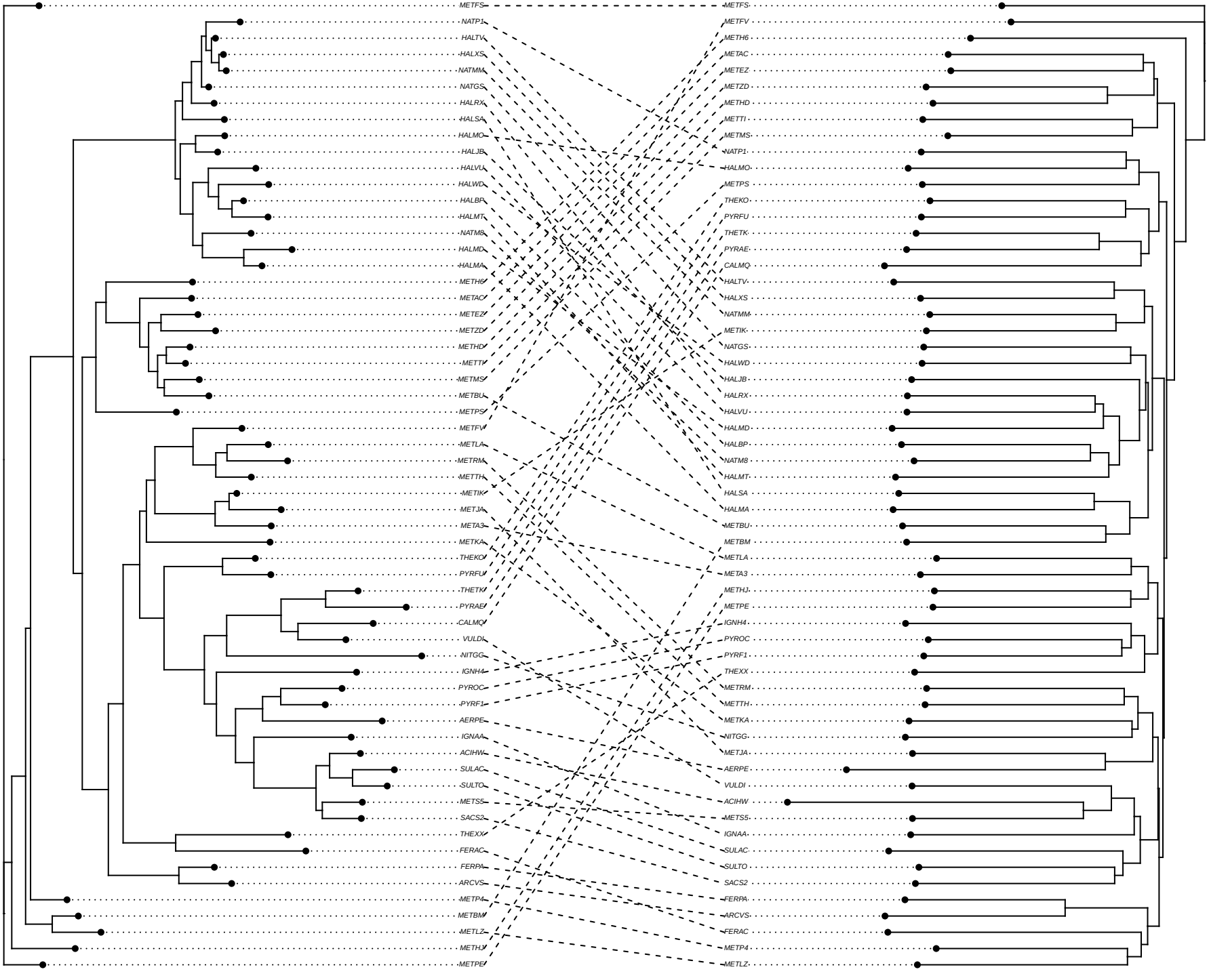
# IQ-TREE vs CMAPLE



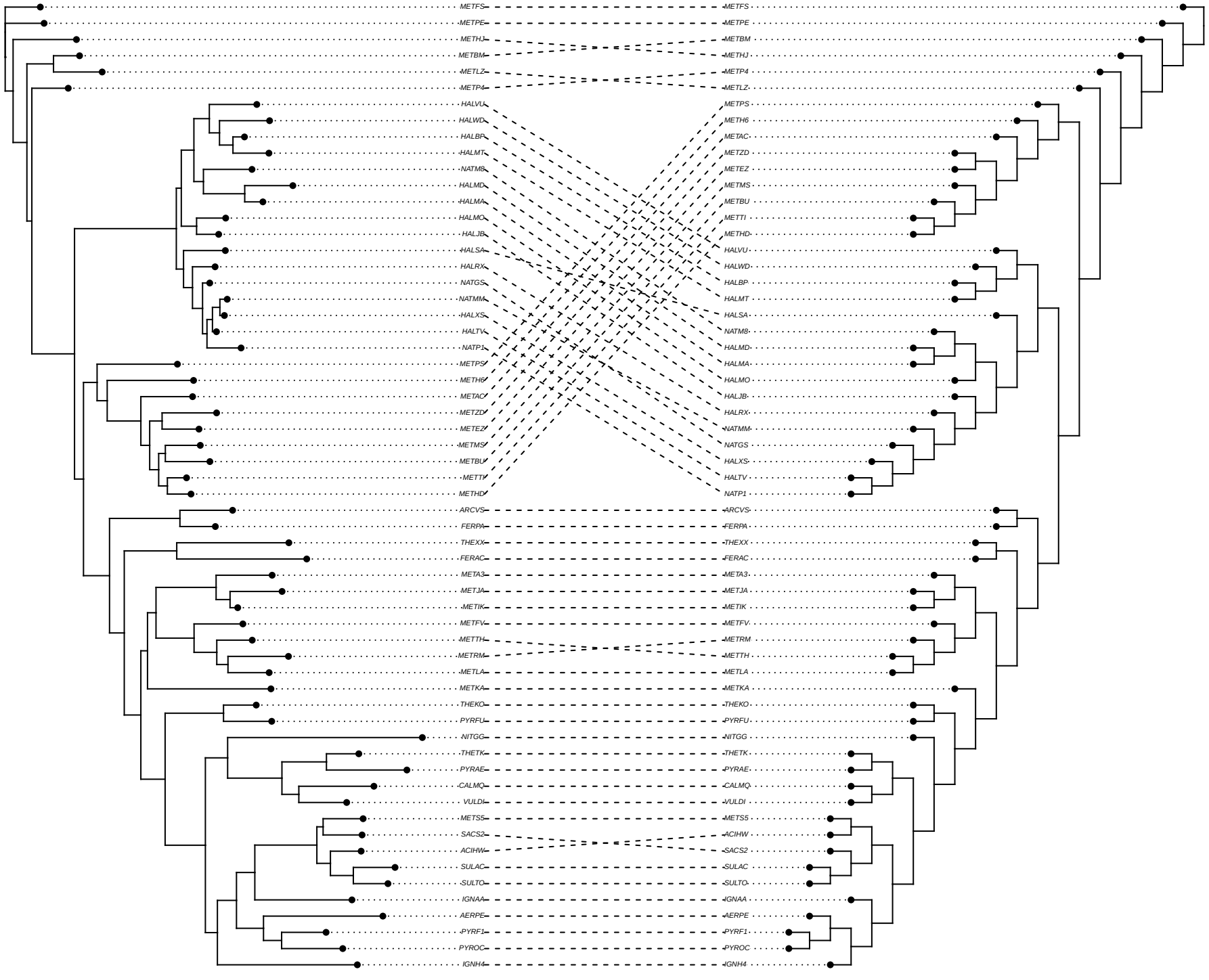
# IQ-TREE vs MSA-Pattention



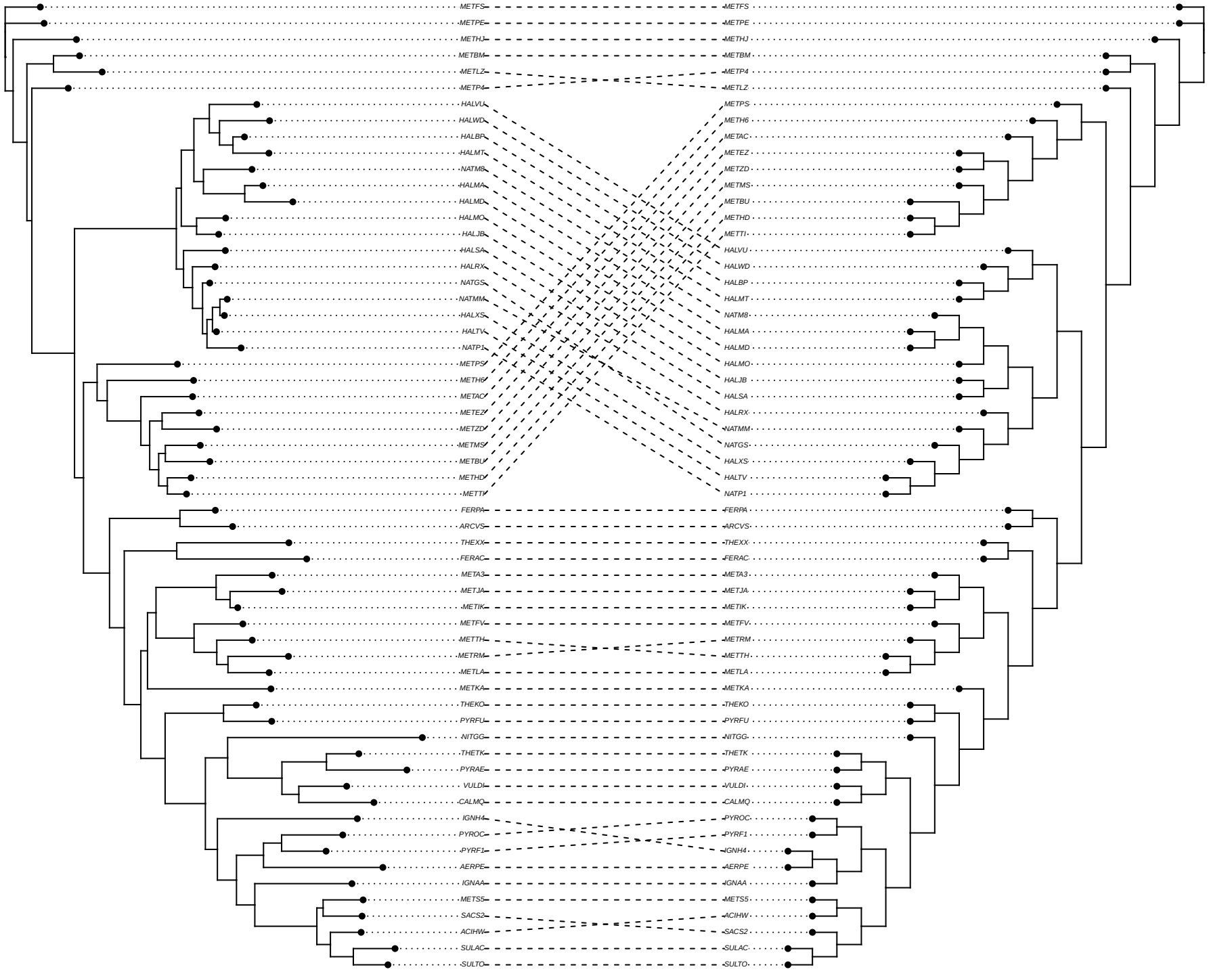
# IQ-TREE vs MSA-T embedding

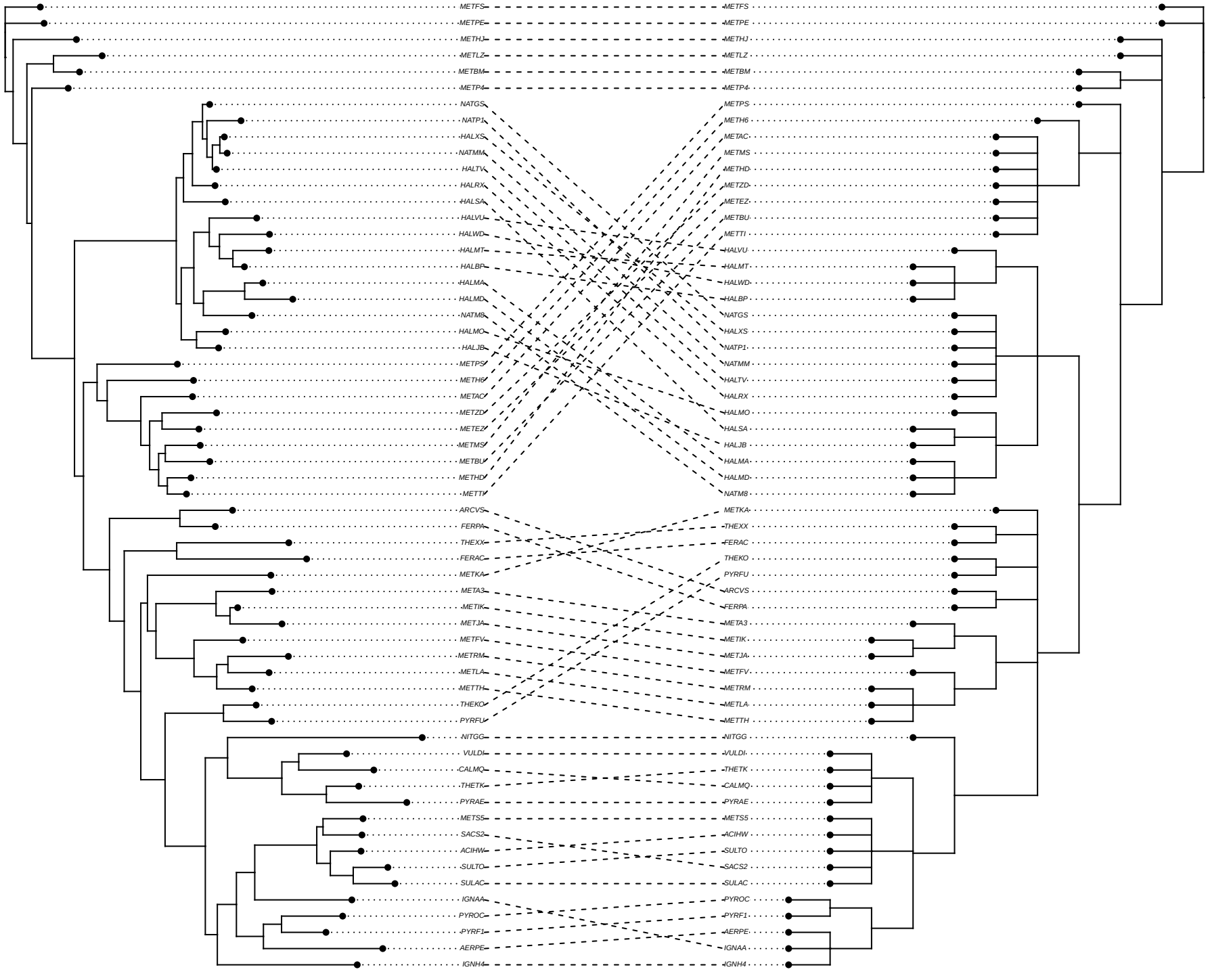


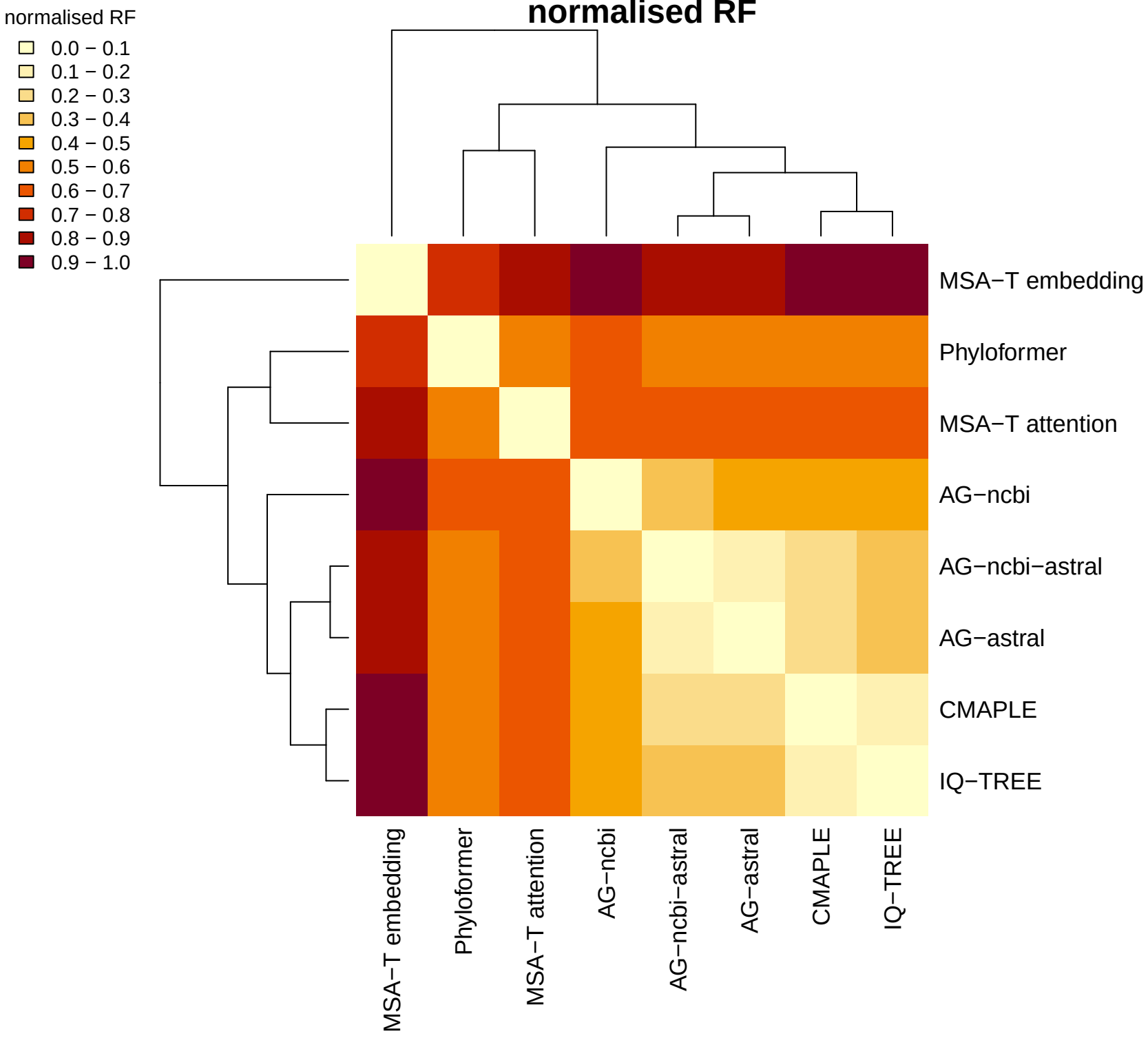
# IQ-TREE vs AG-astral

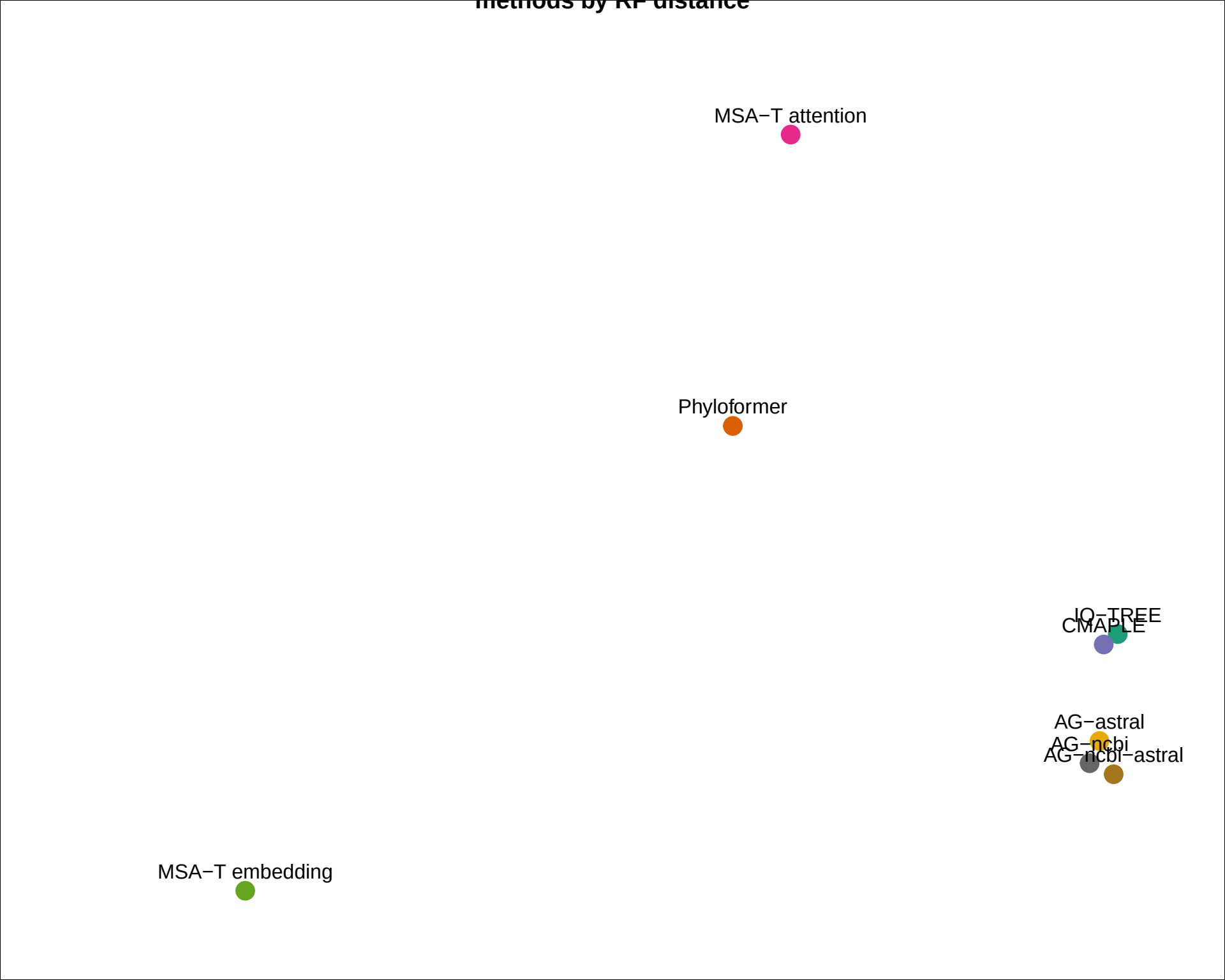


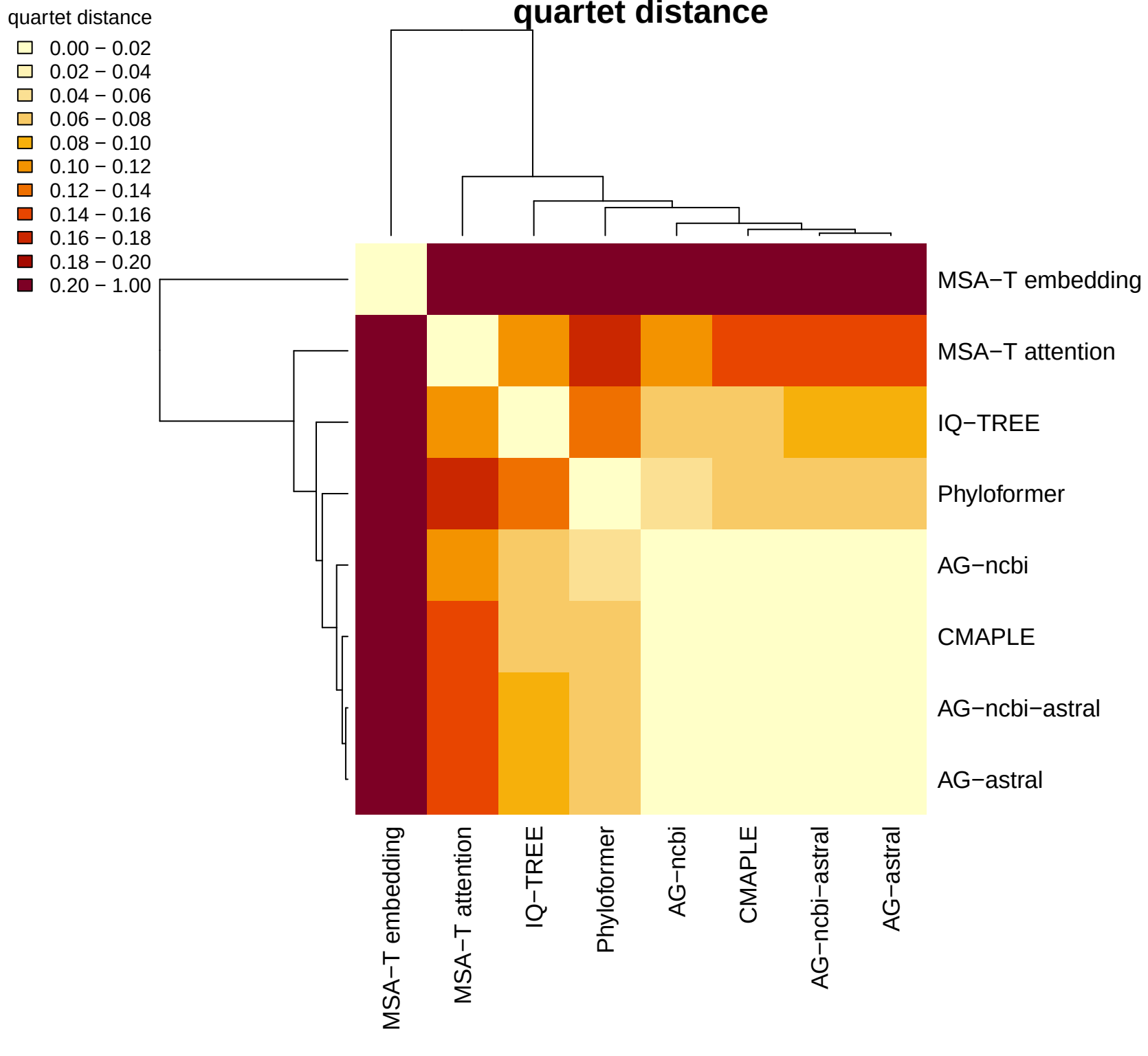
# IQ-TREE vs AG-ncbi-astral

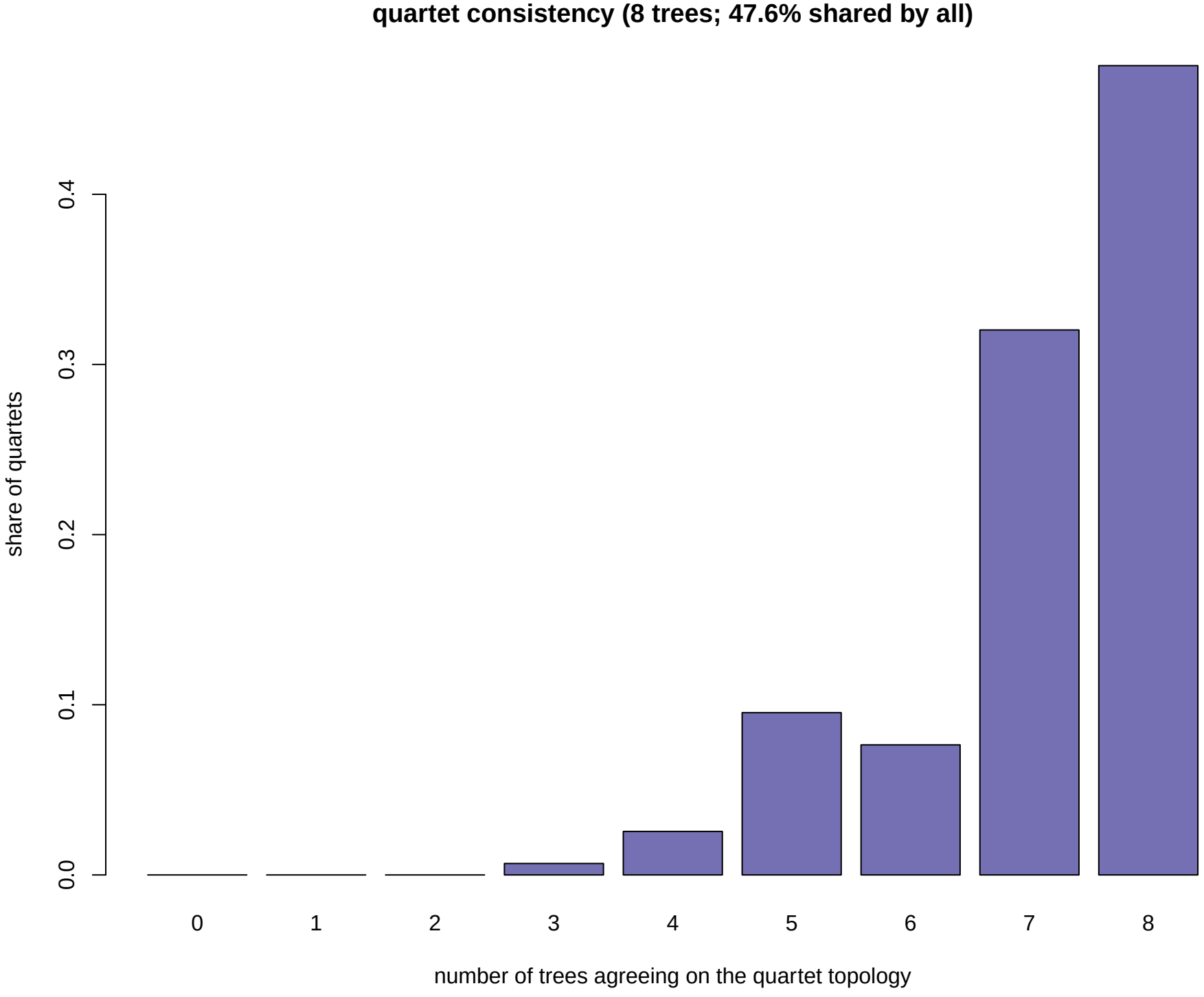


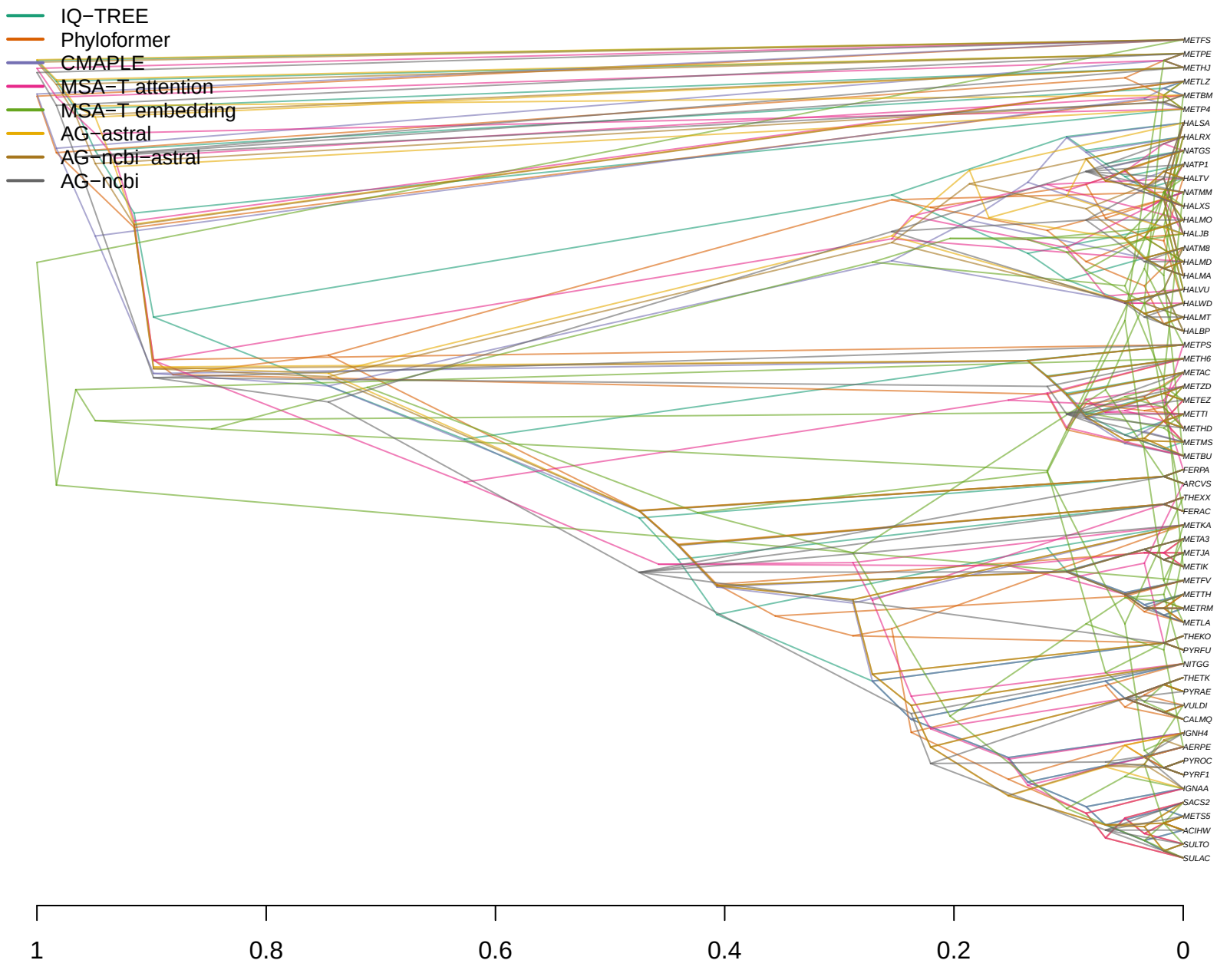

IQ−TREE vs AG−ncbi
METFS
METPE
METHJ
METLZ
METBM
METP4
NATGS
NATP1
HALXS
NATMM
HALTV
HALRX
HALSA
HALVU
HALWD
HALMT
HALBP
HALMA
HALMD
NATM8
HALMO
HALJB
METPS
METH6
METAC
METZD
METEZ
METMS
METBU
METHD
METTI
ARCVS
FERPA
THEXX
FERAC
METKA
META3
METIK
METJA
METFV
METRM
METLA
METTH
THEKO
PYRFU
NITGG
VULDI
CALMQ
THETK
PYRAE
METS5
SACS2
ACIHW
SULTO
SULAC
IGNAA
PYROC
PYRF1
AERPE
IGNH4

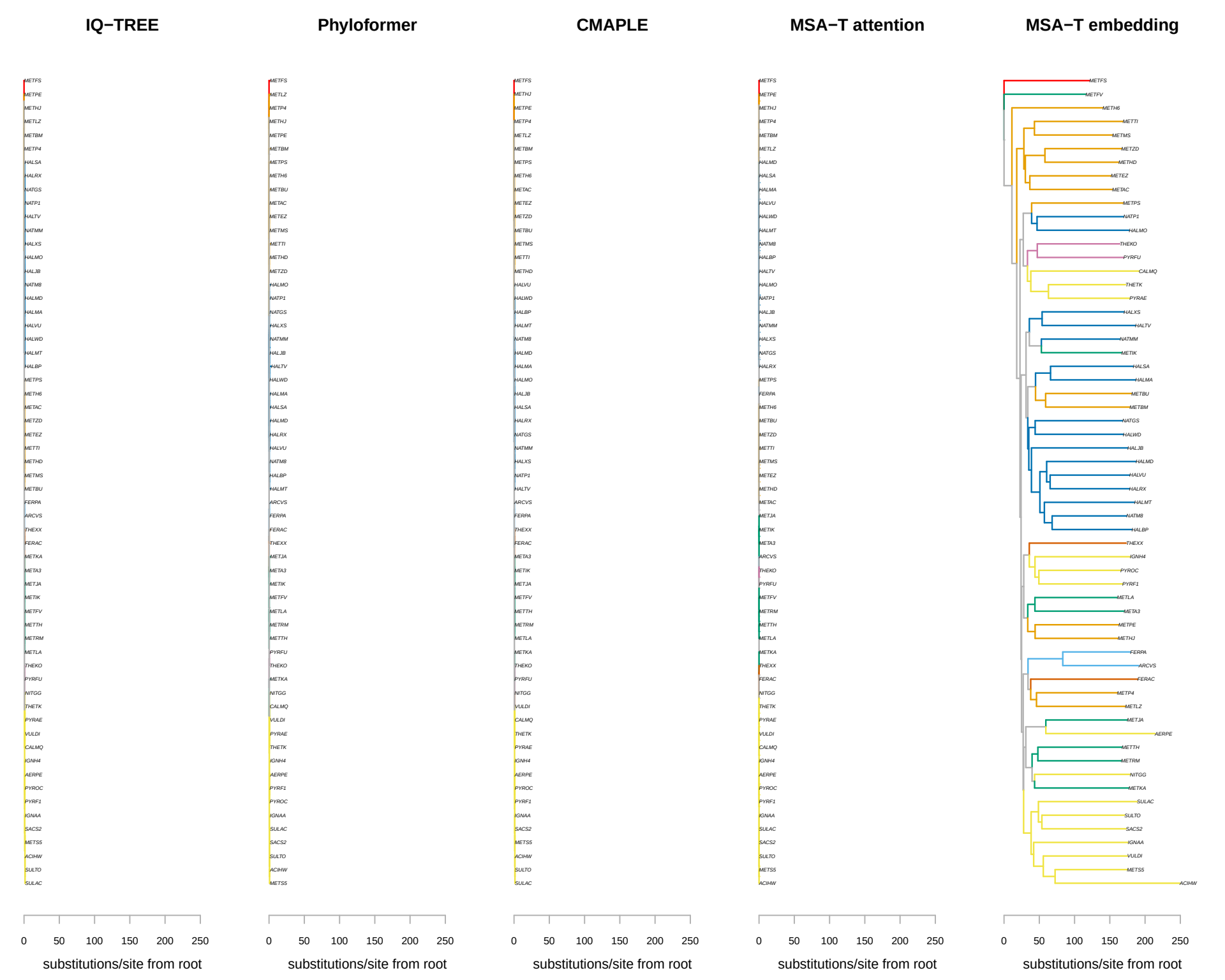

normalised RF
normalised RF
0.0 − 0.1
0.1 − 0.2
0.2 − 0.3
0.3 − 0.4
0.4 − 0.5
0.5 − 0.6
0.6 − 0.7
0.7 − 0.8
0.8 − 0.9
0.9 − 1.0
MSA−T embedding
Phyloformer
MSA−T attention
AG−ncbi
AG−ncbi−astral
AG−astral
CMAPLE
IQ−TREE
MSA−T embedding
Phyloformer
MSA−T attention
AG−ncbi
AG−ncbi−astral
AG−astral
CMAPLE
IQ−TREE


methods by RF distance
MSA−T attention
Phyloformer
IQ−TREE
CMAPLE
AG−astral
AG−ncbi
AG−ncbi−astral
MSA−T embedding

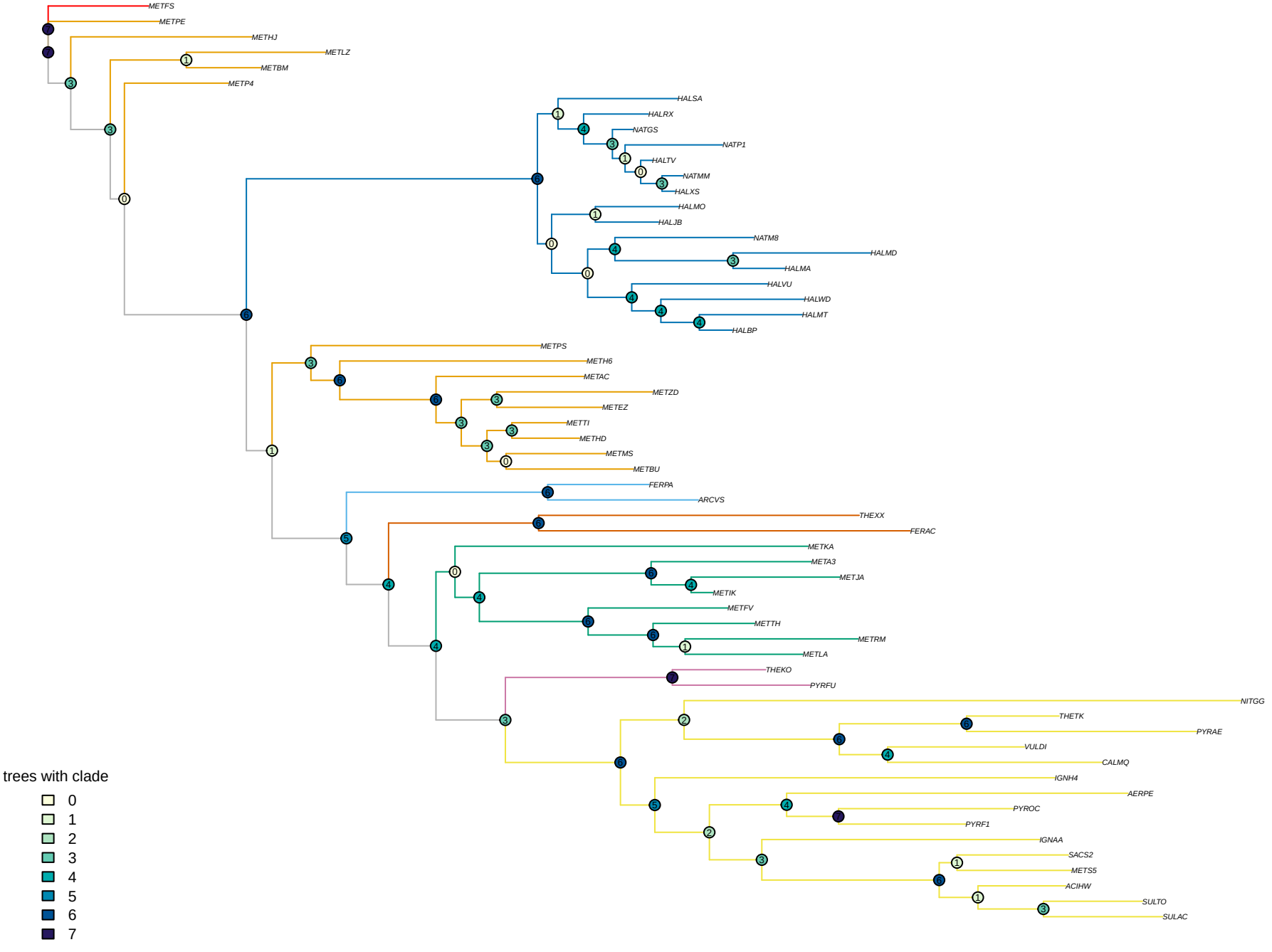

quartet distance
quartet distance
0.00 − 0.02
0.02 − 0.04
0.04 − 0.06
0.06 − 0.08
0.08 − 0.10
0.10 − 0.12
0.12 − 0.14
0.14 − 0.16
0.16 − 0.18
0.18 − 0.20
0.20 − 1.00
MSA−T embedding
MSA−T attention
IQ−TREE
Phyloformer
AG−ncbi
CMAPLE
AG−ncbi−astral
AG−astral
MSA−T embedding
MSA−T attention
IQ−TREE
Phyloformer
AG−ncbi
CMAPLE
AG−ncbi−astral
AG−astral


quartet consistency (8 trees; 47.6% shared by all)
share of quartets
0.0
0.1
0.2
0.3
0.4
0
1
2
3
4
5
6
7
8
number of trees agreeing on the quartet topology

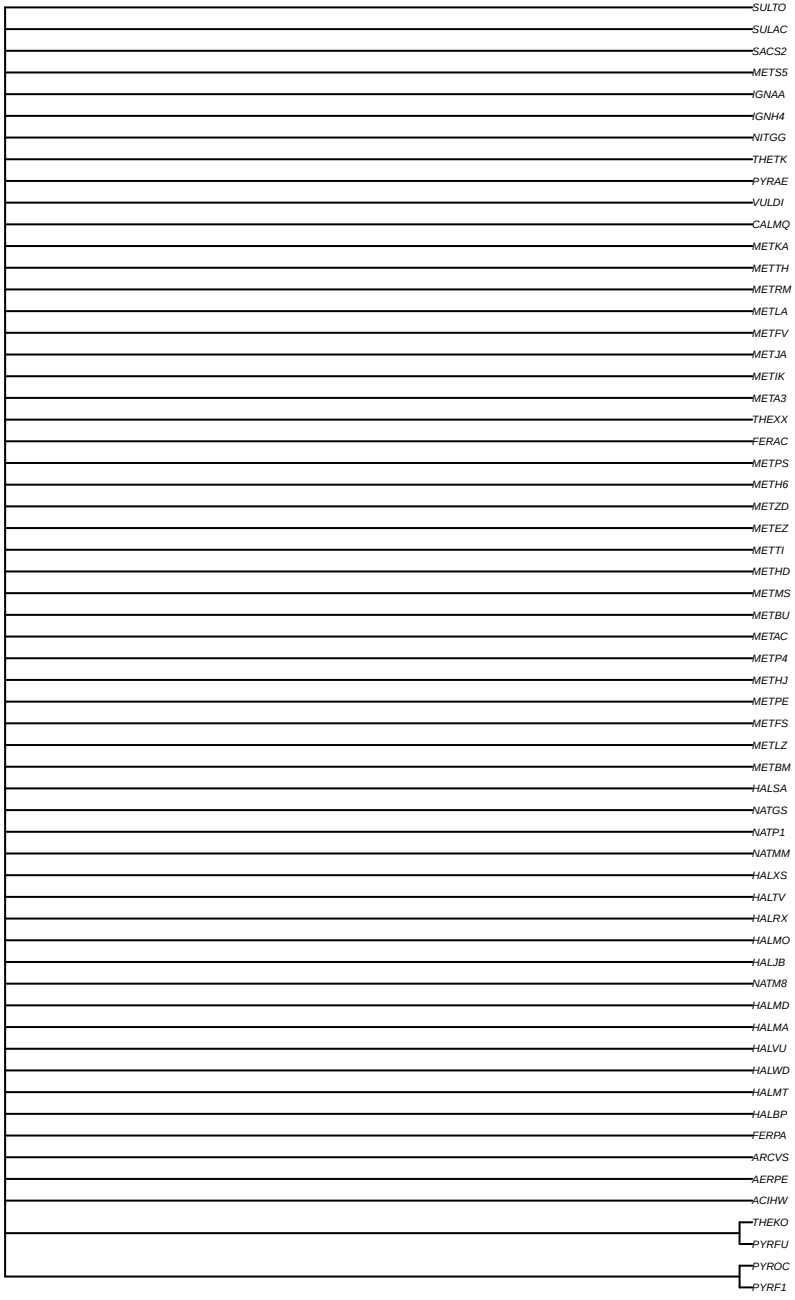

densitree: topology only (branch lengths ignored)
IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
AG−astral
AG−ncbi−astral
AG−ncbi
METFS
METPE
METHJ
METLZ
METBM
METP4
HALSA
HALRX
NATGS
NATP1
HALTV
NATMM
HALXS
HALMO
HALJB
NATM8
HALMD
HALMA
HALVU
HALWD
HALMT
HALBP
METPS
METH6
METAC
METZD
METEZ
METTI
METHD
METMS
METBU
FERPA
ARCVS
THEXX
FERAC
METKA
META3
METJA
METIK
METFV
METTH
METRM
METLA
THEKO
PYRFU
NITGG
THETK
PYRAE
VULDI
CALMQ
IGNH4
AERPE
PYROC
PYRF1
IGNAA
SACS2
METS5
ACIHW
SULTO
SULAC
1
0.8
0.6
0.4
0.2
0


IQ−TREE
Phyloformer
CMAPLE
MSA−T attention
MSA−T embedding
METFS
METPE
METHJ
METLZ
METBM
METP4
HALSA
HALRX
NATGS
NATP1
HALTV
NATMM
HALXS
HALMO
HALJB
NATM8
HALMD
HALMA
HALVU
HALWD
HALMT
HALBP
METPS
METH6
METAC
METZD
METEZ
METTI
METHD
METMS
METBU
FERPA
ARCVS
THEXX
FERAC
METKA
META3
METJA
METIK
METFV
METTH
METRM
METLA
THEKO
PYRFU
NITGG
THETK
PYRAE
VULDI
CALMQ
IGNH4
AERPE
PYROC
PYRF1
IGNAA
SACS2
METS5
ACIHW
SULTO
SULAC
METFV
METLZ
METP4
METHJ
METPE
METBM
METPS
METH6
METBU
METAC
METEZ
METZD
METMS
METTI
METHD
METZD
HALMO
NATP1
NATGS
HALXS
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HALJB
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HALMA
HALSA
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NATM8
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FERPA
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THEXX
METJA
META3
METIK
METFV
METLA
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METTH
PYRFU
THEKO
METKA
NITGG
CALMQ
VULDI
PYRAE
THETK
IGNH4
AERPE
PYRF1
PYROC
IGNAA
SULAC
SACS2
SULTO
ACIHW
METS5
METFS
METHJ
METPE
METP4
METLZ
METBM
METLZ
HALMD
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HALMT
NATM8
HALBP
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HALMO
NATP1
HALJB
NATMM
HALXS
NATGS
HALRX
METPS
FERPA
METH6
METBU
METZD
METTI
METMS
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PYRAE
VULDI
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IGNH4
AERPE
PYROC
PYRF1
IGNAA
SULAC
SACS2
SULTO
METS5
ACIHW
METFS
METFV
METH6
METTI
METMS
METZD
METHD
METEZ
METAC
METPS
NATP1
HALMO
THEKO
PYRFU
CALMQ
THETK
PYRAE
HALXS
HALTV
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HALRX
HALMT
NATM8
HALBP
THEXX
IGNH4
PYROC
PYRF1
METLA
META3
METPE
METHJ
FERPA
ARCVS
FERAC
METP4
METLZ
METJA
AERPE
METTH
METRM
NITGG
METKA
SULAC
SULTO
SACS2
IGNAA
VULDI
METS5
ACIHW
0
50
100
150
200
250
substitutions/site from root

# IQ−TREE clades: number of the other 7 trees that contain each clade



## strict




## majority rule

